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# Designing the SmartVac Model-Based Reflex Agent

**Executive Summary:** This report designs *SmartVac*, a home-cleaning robot agent built as a **model-based reflex agent**. The agent operates in a dynamic, partially observable home environment (rooms, furniture, pets). It uses onboard sensors (bump, cliff, dirt, battery, etc.) to update an internal state (map and battery level) and applies condition-action rules (reflexes) to decide actions. Its performance is measured by cleaning efficiency (area cleaned, energy use, and safety). We assume a home vacuum-cleaner domain. The PEAS (Performance, Environment, Actuators, Sensors) of SmartVac is detailed below, with key trade-offs in sensor/actuator choices. Figures and tables illustrate the agent architecture, data flow, and component comparisons. We discuss limitations of this simple design (limited planning, possible loops) and suggest improvements (adding mapping, learning, richer sensors)

## 1. Agent Type Selection and Naming

We choose a **model-based reflex agent** for SmartVac. Simple reflex agents act only on the current percept (e.g. a thermostat that switches on/off by temperature) and require a fully observable environment. In contrast, a home environment is **partially observable and dynamic** (furniture, pets move), so SmartVac must remember some state (e.g. previously cleaned areas, orientation). A model-based agent maintains an internal model of the world to handle hidden or changing information. Thus it updates its state from sensors and then applies reflex rules.

## 2. Graphical Representation (Agent Architecture)

*The SmartVac home-cleaning agent. It perceives the environment (house rooms) via sensors (e.g. bump and cliff detectors, dirt sensor) and acts via its actuators (wheels for movement, brushes/suction for cleaning).*

Figure 1 shows SmartVac in a home setting. SmartVac's architecture follows the classic *perception–action loop*: The **Environment** (home interior) provides percepts to **Sensors**, which feed data to the **Agent Program**. The agent updates its internal state and decides an **Action**, executed via **Actuators** on the environment.

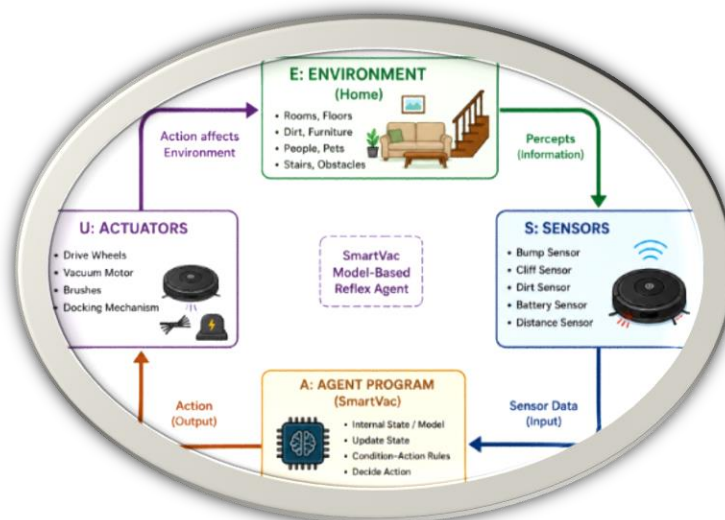
### Graph

E [Environment (Home)] --> S[Sensors]

S --> A [Agent Program (SmartVac)]

A --> U [Actuators]

U --> E



### 3. Agent Formulation

**Environment:** A home (indoor) environment with rooms, floors (carpet, wood), furniture, obstacles (toys, cables), and dynamic elements (people, pets). The environment is *partially observable* and *stochastic*: SmartVac cannot see the whole house at once and furniture/pets may move. It must adapt to unknown layouts.

**Sensors:** SmartVac includes multiple sensors.

**Bump/collision sensors** to detect obstacles.

**Cliff/edge sensors** to detect stairs or drop-offs.

**Dirt sensor** (e.g. optical or acoustic) to detect unclean patches.

**Battery level sensor** to monitor remaining power.

**Actuators:** The actuators allow movement and cleaning

**Drive wheels** (e.g. differential drive) to move/steer SmartVac.

**Vacuum motor and brushes** to suck up dirt.

**Brush wheels** or turret to orient the vacuum.

**Docking actuator** (magnet or guide) to return to a charging station.

### 4. PEAS Evaluation

The PEAS framework specifies the agent's task environment:

| Aspect                 | SmartVac Specification                                                                                                                                             |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Performance</b> (P) | Cleanliness (area cleaned), coverage of floors; energy efficiency (battery usage); time to finish; safety (no falls or damage).                                    |
| <b>Environment</b> (E) | Home interior: multi-room floor plan with static (walls, furniture) and dynamic (people, pets) elements. Floors of various types; lighting levels; stairs present. |
| <b>Actuators</b> (A)   | Drive wheels (forward/back, turn); brush rotation and vacuum suction; docking for charging. These change position and clean debris.                                |
| <b>Sensors</b> (S)     | Bump/collision sensors; cliff (drop-off) detectors; dirt sensor; battery gauge. Optionally, distance sensors (IR/ultrasonic) or camera for navigation.             |

## Sensor and Actuator Trade-offs

| Sensor             | Function                    | Pros                                           | Cons                                                                |
|--------------------|-----------------------------|------------------------------------------------|---------------------------------------------------------------------|
| Bump Switch        | Detect physical collisions  | Simple, very reliable on contact               | Only triggers on impact; can miss soft obstacles (e.g. curtains)    |
| Infrared (IR)      | Obstacle/distance detection | Non-contact, low-cost, detects walls/furniture | Limited range; affected by lighting and reflective surfaces         |
| Camera/Camera+SLAM | Visual navigation/mapping   | Rich information, can localize/plan            | High compute cost; complex to implement; may require good lighting  |
| Dirt Sensor        | Detect dirt concentration   | Focus cleaning effort, efficient               | Measures only local dust; may be fooled by non-dirt particles       |
| Cliff Sensor (IR)  | Detect drop-offs (stairs)   | Essential for safety (avoiding falls)          | Only senses edges beneath robot; calibration needed for floor types |
| Battery Sensor     | Monitor charge level        | Ensures return to charge                       | Adds hardware complexity; essential for runtime management          |

| Actuator             | Purpose                  | Pros                                       | Cons                                                         |
|----------------------|--------------------------|--------------------------------------------|--------------------------------------------------------------|
| Differential Wheels  | Mobility (forward, turn) | Simple control, common hardware            | Limited maneuverability in tight spots; non-holonomic motion |
| Vacuum Motor & Brush | Dirt removal             | Effective cleaning; continuous dirt intake | Power-hungry; requires maintenance (brush wear)              |
| Rotating Head/Servo  | Change orientation       | Allows turning on the spot                 | Additional mechanical complexity; potential failure point    |
| Docking Mechanism    | Automated recharging     | Self-sufficiency (no manual charging)      | Requires precise positioning; may fail if alignment off      |

## Limitations:

A model-based reflex agent is fairly simple and may behave sub optimally. Because decisions are rule-based and myopic, SmartVac can get trapped in loops or miss optimal paths

